

AICE2005 - Systematic Design

Gathering and Validating Requirements

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Week Overview

Week 2 | Analysing systems: requirements

Lecture 1: Functional and Non-Functional Requirements

What must the system do, and how well must it do it?

Lecture 2: Gathering and Validating Requirements

How do we find out what stakeholders need, and check we got it right?

Lecture 3: Use Cases and Personas

How do we capture users and their goals as scenarios?

Recap: Lecture 1

- **FR vs NFR:** what the system does vs how well it does it
- **Measurable NFRs:** value, unit, condition, proportion and method
- **Precise wording:** five questions, logic, EARS templates
- **Trade-offs:** NFRs conflict and are allocated across components

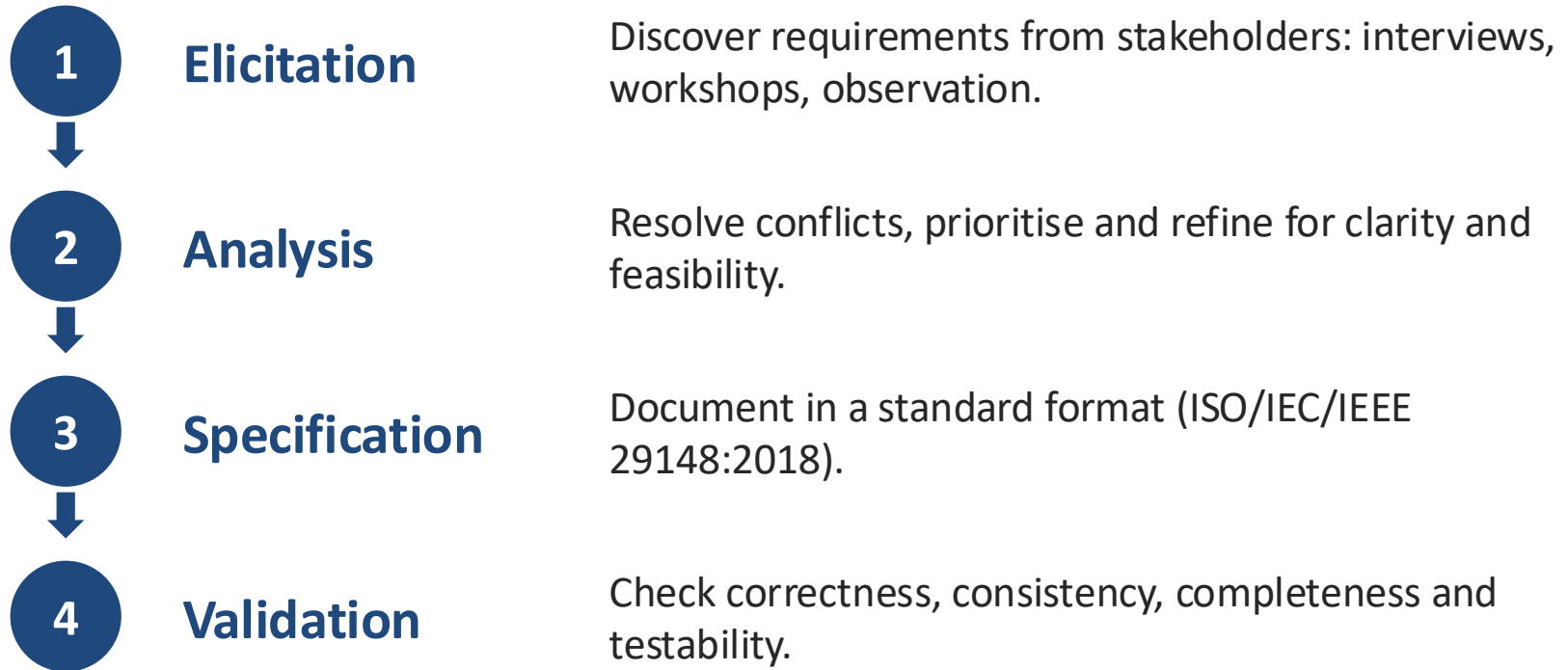
Lecture 2 Overview

- The requirements lifecycle
- Stakeholders and elicitation techniques
- Validating requirements
- Conflicts, trade-offs, security and ethics
- Case study: a child-monitoring toy duck

Learning Objectives:

- Apply appropriate techniques to gather user and stakeholder requirements
- Validate requirements for completeness, consistency and feasibility
- Evaluate trade-offs in complex system requirements
- Identify challenges in multi-stakeholder systems and propose resolution strategies

The Requirements Lifecycle



Management: track and update requirements throughout the system lifecycle.

Stakeholders and Sources

Stakeholders

- Users
- Project sponsors
- Regulatory bodies
- Developers and testers
- Maintainers
- Adjacent systems (yes, systems too)

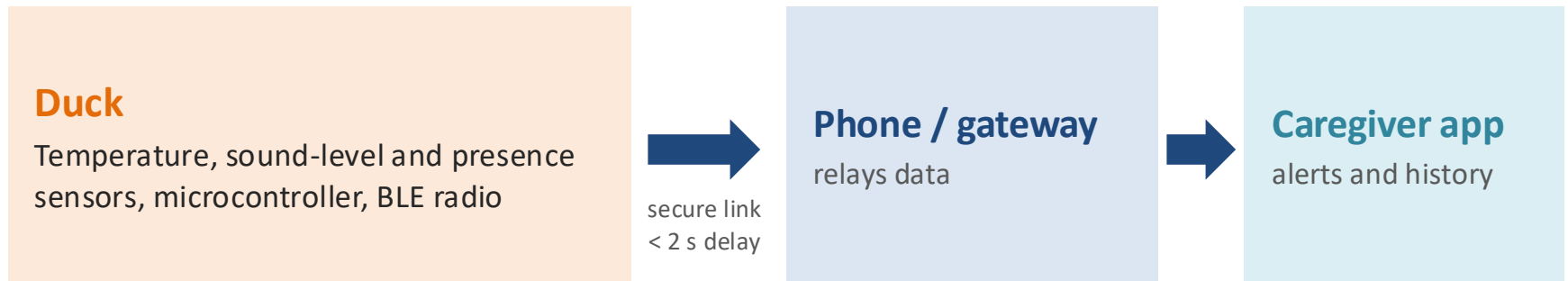
Sources of requirements

- User interviews and observation
- Existing systems and documentation
- Standards and regulations
- Competitive benchmarking
- Internal expertise and R&D insights

**Identify stakeholders early: a missed stakeholder is a missed requirement.
Stakeholders are more than users.**

Case Study: Child-Monitoring Duck

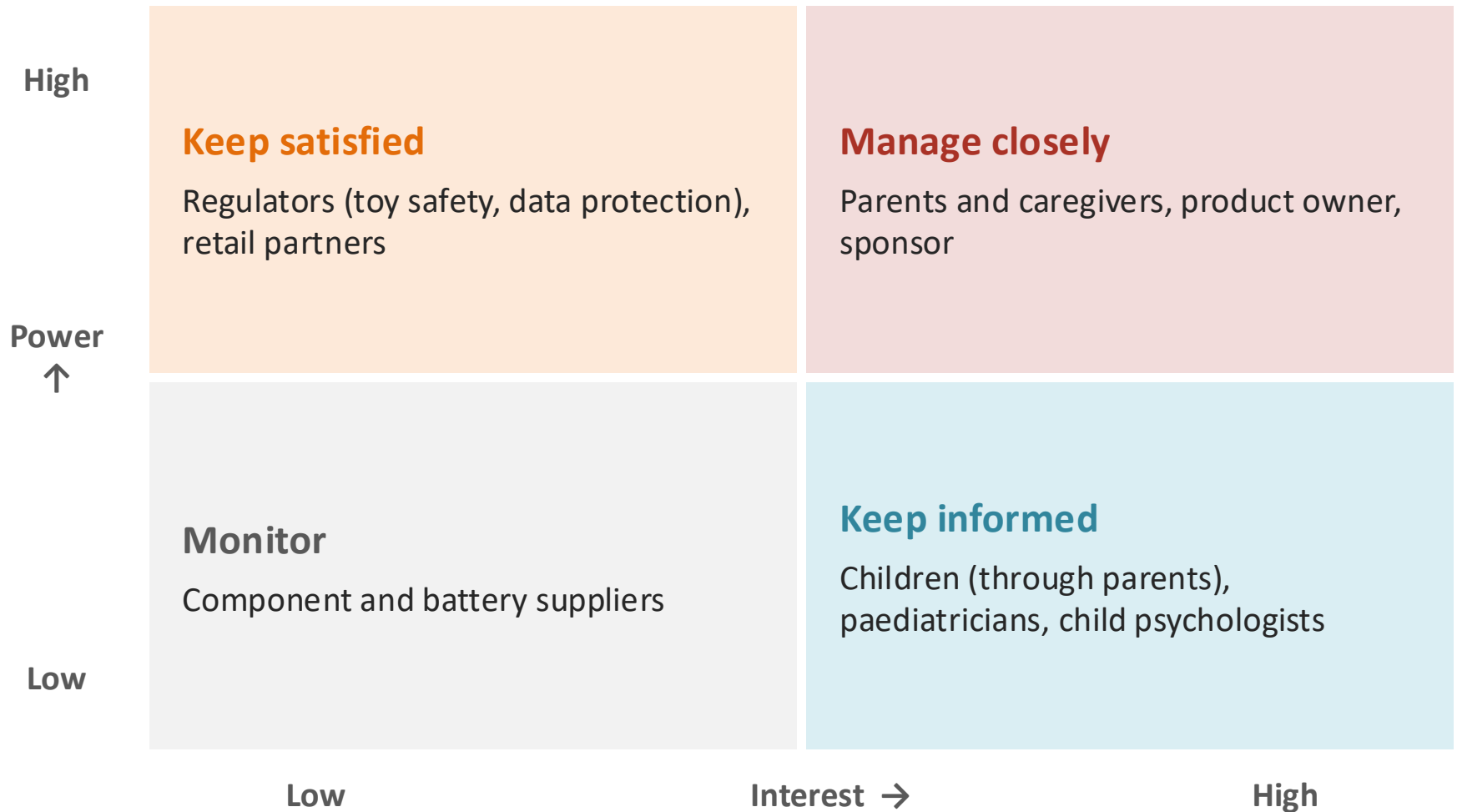
Design a child-monitoring system embedded in a toy duck, used by caregivers for indoor safety monitoring.



System constraints

- Must resemble a rubber duck in size, shape and texture.
- Must be safe and non-intrusive for children aged 2 to 5.
- Must monitor room temperature and sound levels, and detect presence.
- All data must reach the caregiver app securely with minimal latency (< 2 s).

Mapping Stakeholders: Power and Interest



Children have limited decision-making power but are primary users and affected stakeholders; their requirements should therefore be actively represented.

Elicitation Techniques

Technique	Best for	Watch out for
Interviews	Deep insight, implicit concerns	Interviewer bias, small samples
Questionnaires and surveys	Many or dispersed users, trends	Shallow answers, badly worded questions
Workshops and focus groups	Brainstorming, resolving conflicts	Dominant voices
Observation (ethnography)	Tacit workflows, real context	Time-consuming, observer effect
Document and system analysis	Legacy systems, standards	Outdated documentation
Prototyping	Feasibility, concrete reactions	Users fixate on the look
User stories and use cases	Actor-system interactions, agile	Quality requirements get missed

Asking Good Questions

Ask

- "Walk me through the last time you..."
- "What happens when it goes wrong?"
- "What must it never do?"
- "Who else should we talk to?"

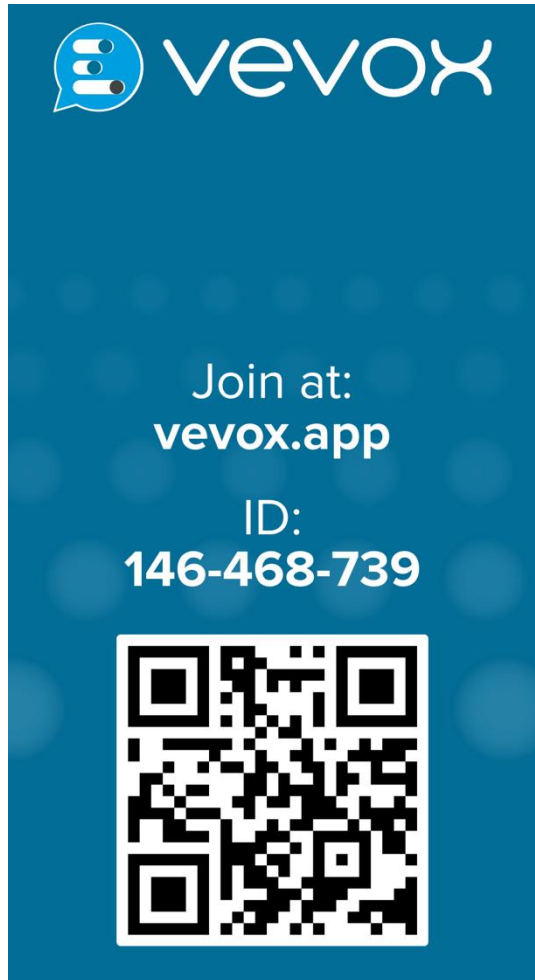
Avoid

- **Leading:** "Don't you find the app confusing?"
- **Double-barrelled:** "Is it fast and easy?"
- **Hypothetical:** "Would you use it?" (ask about past behaviour instead)

People cannot tell you what they do without noticing. Combine interviews with observation, and use closed questions to quantify what open questions reveal.

Let's think and reflect

Exercise: Which technique?



Pick the best technique on Vevox

Interview, survey, workshop, observation, document analysis or prototype.

1. See how nurses really use a ward tablet while walking.
2. Get quick feedback from 800 students on several campuses.
3. Resolve a clash between the sensor team and the app team.
4. Find out what an undocumented legacy system really does.
5. Test whether users grasp a new dashboard idea before building it.
6. Explore a lecturer's worries about automated marking.

Validating Requirements

Validation

Are we building the right system?

Checks requirements against real stakeholder needs.

Verification

Are we building the system right?

Checks the design and code against the requirements.

Requirements must be:

Correct: reflect true stakeholder intent

Consistent: no internal contradictions

Complete: cover all relevant scenarios

Feasible: achievable within constraints

Verifiable: can be tested or demonstrated

Validation catches wrong requirements before they are built, when the cost of change is lowest.

Validation Techniques

Technique	What it checks	Example
Requirement reviews	Ambiguity, omissions, inconsistency	Walkthrough with reader, reviewer, moderator
Simulation and prototyping	Can it be met in real conditions?	FEA for hardware; wireframes for software
Test-case derivation	Is it verifiable?	"The arm must lift 1 kg in 5 s in 90% of trials"
Traceability (RTM)	Is every requirement covered?	Requirement to design to test; impact analysis
Feasibility checks	Time, cost, technical limits	Are interoperable interfaces available?
Stakeholder review sessions	Are human needs kept?	Usability and accessibility walkthroughs
Requirement modelling	Does the model satisfy properties?	SysML, model checking, formal methods

Conflicts and Prioritisation

Must

Dashboard shows room temperature

Should

Occupancy history for the last week

Could

Study-tip suggestions

Won't (this release)

Voice assistant

MoSCoW: Must, Should, Could, Won't

Managing conflicts

- Rank with stakeholders, then decide and record the reason.
- Quantify the conflict: turn opinions into numbers.
- Model the system (system maps, giga-mapping) to see conflicts and leverage points.

Duck: How We Gather and Validate

Gathering

- Stakeholder interviews with parents, paediatricians and child psychologists.
- Competitive analysis of baby monitors and smart toys.
- Meta-models to explore trade-offs between sensor placement, battery capacity and toy softness.

Validating

- **Functional:** embedded system simulation (microcontroller, BLE, sensor modules).
- **Safety:** compliance check against toy safety standards (e.g. EN 71).
- **Stakeholder:** physical prototypes tested in supervised play environments.

Each validation type answers a different question: does it work, is it safe, do people accept it?

Security and Ethics Are Requirements Too

Data in motion

- Duck-to-app link shall be encrypted and paired.
- Cloud traffic shall use TLS 1.3.

Data at rest

- Stored data shall be encrypted (AES-256).
- Access limited to the caregiver's account.

Ethics and privacy

- Sound level only: no audio is recorded.
- Data minimised; deleted after 30 days.

Ask about every piece of data:

- Who can see it, and why do we collect it at all?
- How long do we keep it, and who can delete it?
- Young children cannot consent: parents do. UK GDPR gives children's data special protection.

Summary and What's Next

- **Lifecycle:** elicit, analyse, specify, validate, manage: a loop
- **Stakeholders:** identify early, map power and interest
- **Techniques:** match the technique to what you need to learn
- **Validation:** right system, then system right
- **Trade-offs and ethics:** prioritise, record decisions, protect data

Let's read and explore

- ISO/IEC/IEEE 29148:2018. Requirements engineering.
<https://www.iso.org/standard/72089.html>
- Sommerville, I. Software Engineering, Ch. 4 Requirements engineering.
https://www.oreilly.com/library/view/software-engineering-10th/9780137586691/xhtml/fileP700049592600000000000000000000AD0.xhtml?utm_source=chatgpt.com
- Robertson, S. and Robertson, J. Mastering the Requirements Process.
<https://www.oreilly.com/library/view/-/9780132942850>.
- Errikos, S. et al. Managing Ethical Requirements Elicitation
https://link.springer.com/chapter/10.1007/978-3-031-42307-9_19

Next Lecture: Use Cases and Personas: how do we capture users and their goals as scenarios?